

Curvature Polarization Transport Gravity

A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions

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Abstract

Curvature Polarization Transport Gravity (CPTG) is a unified geometric framework in which General Relativity (GR), curvature polarization, and curvature transport operate as a coupled gravitational hierarchy. In the gravitational sector, ordinary physical stress–energy sources curvature through a parent action whose Einstein term remains present throughout, while the exact GR field equation is recovered continuously when the polarization–transport response becomes negligible. Bound-system, merger, compact, and cosmological equations are registered symmetry-, scale-, or state-dependent realizations of that common architecture. Primordial nuclear reactions are not governed by the gravitational field equation; they provide a conserved reaction-state realization of the same polarization–transport organization, propagated by native nuclear dynamics.

A central structural result is Structural Mode N , a dimensionless post-solution measure of how the solved CPTG response is organized. In the galaxy reduction, the structural acceleration scale $a_\star$ is inferred from resolved kinematic and baryonic source data, the associated R_c is derived, and only then is N evaluated from the retained curvature structure. The Curvature Structure Mode Index (CSMI) converts that solved coordinate into a structural catalog without using Hubble type or visual morphology as an object-level input. The resulting classification robustly orders non-input galaxy structure and is empirically distinct from $a_\star$. The same framework also contains directional merger transport and finite-curvature compact admissibility, while galaxy clusters remain within the bound-system structural hierarchy.

Cosmologically, CPTG contains one native π -branch. Acoustic, luminosity, cosmic microwave background (CMB), growth, and lensing quantities are downstream projections or comparison coordinates of that native geometry. In the primordial nuclear sector, exact stoichiometry gives three ordered transport directions plus one charge-constrained polarization direction, producing an exact rank-four light-sector reaction-source coordinate whose dynamical pushforward is supplied by native reaction networks.

1 Introduction

Galaxy rotation curves, cluster lensing, compact objects, cosmological expansion, cosmic microwave background (CMB) observables, and primordial nuclear reactions are normally analyzed with different variables and numerical tools [2, 3, 5, 7, 26]. CPTG asks whether part of that apparent separation reflects different reductions or projections of one geometric response rather than independent hidden sectors.

In the gravitational theory, ordinary physical stress–energy is the material source. The additional response is geometric: curvature polarization describes how the sourced field reorganizes nonlinearly, while curvature transport describes how that organized response persists, redistributes, or becomes directionally displaced. The bound-system reductions use resolved baryonic matter as their material input, while the parent stress–energy tensor retains its ordinary broader role for radiation, pressure, internal energy, neutrinos, electromagnetic fields, and other physically present non-dark-sector sources.

Different physical regimes therefore retain different variables without becoming different theories. A quasi-static galaxy admits a radial response with a structural transport remnant; a galaxy cluster remains an extended bound system whose baryonic source distribution defines the local source geometry, while rotational, dispersion, thermal, or mixed support provides observational constraints on the resulting structural response; a dissociative merger retains directional transport explicitly; a compact object probes the extreme high-gradient response; and cosmology is described by one native π -branch whose observational channels are projected downstream. In primordial nuclear physics, the same polarization–transport organization appears as a conserved reaction-state geometry that is propagated by the native reaction network rather than by the gravitational field equation [22]. In disk galaxies this coupled hierarchy has a direct dynamical interpretation. The higher-acceleration inner response remains close to the directly baryon-sourced limit, while toward the lower-acceleration outer galaxy the polarization and residual transport contributions become increasingly important in supporting the observed rotational kinematics. CPTG therefore seeks to account for the outer gravitational response from the observed baryonic source and geometric polarization–transport response rather than by adding a dark-matter halo.

The common theoretical ordering is

- physical source and native geometric state
- ↓ geometric invariants, symmetry, and boundary/state data
- ↓ declared gravitational reduction or native state-space realization
- ↓ polarization–transport dynamics or native dynamical pushforward (1)
- ↓ solved physical state
- ↓ observational or network projection
- reported comparison quantity.

The direction of this hierarchy is part of the theory. A gravitational reduction is declared

before its reduced polarization–transport equations are solved; in the nuclear realization, exact source geometry is fixed before the native dynamical pushforward. Downstream observational coordinates may test a fixed CPTG construction but do not become new native fields, matter components, or independent cosmological branches merely because they are useful for comparison. This is especially important in cosmology, where acoustic, luminosity, CMB, growth, and lensing coordinates remain projections of one native history, and in nuclear physics, where each admissible native reaction network retains its own rates and dynamics [18, 20, 21, 22].

Structural Mode N is a particularly direct consequence of this ordering. In the galaxy realization it is calculated only after the structural acceleration scale a_* and nonlinear field solution have been selected, so it measures the organization of the solved polarization–transport structure rather than supplying an additional fit parameter. The dedicated Structural Mode study establishes the empirical role of this coordinate and the resulting CSMI classification without importing a conventional morphology label into the calculation [12].

Because the maturity of the sectors differs, the paper distinguishes native definitions, exact consequences, reduced physical laws, projected or data-conditioned coordinates, numerical validations, and open extensions. The sections that follow state the gravitational variational foundation, bound and dynamical structure, native cosmology, observational projections, the primordial nuclear state-space realization, and the corresponding failure conditions.

2 Variational Foundations and Theory Hierarchy

CPTG organizes gravitational response through a coupled hierarchy of General Relativity, curvature polarization, and curvature transport. These components are computed differently but operate together as one geometric gravitational framework. Gravitational sectors are registered reductions or realizations of one CPTG gravity-side architecture within their declared symmetry, scale, state, and boundary domains; non-gravitational sectors may realize the same curvature–polarization–transport organization in their own native state spaces.

2.1 Scale-Structured Realization Principle

CPTG is a framework of physical scales. The common theory fixes the source–response ordering, the polarization–transport architecture, the conservation requirements, and the rule that a physical realization is selected before its downstream observable is constructed. The operative reduced law is the law appropriate to the source, physical scale, symmetry, dynamical state, and boundary or state data of that problem.

A galaxy, coherent cluster, anisotropic merger, compact structure, cosmological background, and conserved nuclear reaction state therefore need not share one reduced component equation. Their consistency is determined by derivation from the common CPTG hierarchy and by agreement wherever two reductions possess a genuine common physical limit. If

two realizations A and B overlap on a physical domain $\mathcal{O}$, every quantity Q that has the same physical meaning in both descriptions must satisfy

$$Q_A|_{\mathcal{O}} = Q_B|_{\mathcal{O}}. \quad (2)$$

Equation (2) is the cross-scale matching rule. It requires shared physical content to agree in the regime where both reductions apply. The GR recovery limit, continuity of structural quantities within a bound-system reduction, and inheritance of one cosmological native state by its downstream projections are instances of this same ordering.

2.2 Curvature Polarization and Curvature Transport

The gravitational sector of CPTG is formulated as a nonlinear geometric response theory defined by a single action-based framework. At the parent gravity level the action is

$$S_{\text{CPTG}} = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R + \mathcal{L}_{\text{pol}} + \mathcal{L}_{\text{trans}} + \mathcal{L}_{\text{m}} \right]. \quad (3)$$

Here $\mathcal{L}_{\text{m}}$ contains the ordinary physical stress–energy source. The additional terms are geometric: $\mathcal{L}_{\text{pol}}$ organizes the nonlinear polarization response of curvature to the sourced field, while $\mathcal{L}_{\text{trans}}$ governs directional transport or redistribution of organized curvature structure. Equation (3) states the common variational architecture of the gravitational sector. The explicit sector formulas used below are reduced representatives in the variables appropriate to each physical realization, rather than one scalar expression imposed on every gravitational state. The nuclear reaction-state construction shares the polarization–transport organization but is not governed by Eq. (3) or its gravitational field equation.

For later conservation statements it is useful to define the geometric response tensors directly by functional differentiation. With

$$S_{\text{pol}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{pol}}, \quad S_{\text{trans}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{trans}}, \quad (4)$$

and the matter convention

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{m}}}{\delta g^{\mu\nu}}, \quad (5)$$

the CPTG response tensors are defined by

$$\Pi_{\mu\nu}^{\text{pol}} = \frac{16\pi G}{c^4} \frac{1}{\sqrt{-g}} \frac{\delta S_{\text{pol}}}{\delta g^{\mu\nu}}, \quad (6)$$

$$\Theta_{\mu\nu}^{\text{trans}} = \frac{16\pi G}{c^4} \frac{1}{\sqrt{-g}} \frac{\delta S_{\text{trans}}}{\delta g^{\mu\nu}}. \quad (7)$$

These definitions are exact at the parent variational level even where the component forms of the two tensors have not yet been reduced explicitly for a particular geometry.

Writing the field equation in terms of these variational responses gives

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (8)$$

The right-hand side remains the ordinary physical stress–energy source. The two additional tensors on the left are geometric response terms. They are not dark-sector stress–energy tensors and are not independently assigned matter fluids. Their role is to encode how the curvature response depends on more than the pointwise amplitude of the source field.

This separation fixes an important interpretation of the theory: enhanced gravitational response and additional material source are not synonymous. If a solved CPTG field exceeds the direct baryonic response, that excess is not entered by adding a new contribution to $T_{\mu\nu}$; within the CPTG description it is carried by the geometric polarization–transport response on the left-hand side of Eq. (8). Thus “more gravitational response” does not by itself imply “more gravitating matter”.

The relationship to General Relativity (GR) is explicit. Wherever the polarization–transport response becomes negligible,

$$\Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} \rightarrow 0, \quad (9)$$

Eq. (8) reduces continuously to the Einstein equation with the same material source. This recovery limit is one end of a continuous response hierarchy rather than a boundary between two unrelated theories. Schematically,

$$\begin{aligned} \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} \simeq 0 &\implies \text{GR-dominated regime,} \\ \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} \text{ non-negligible} &\implies \text{polarization–transport-significant regime.} \end{aligned} \quad (10)$$

As the polarization–transport response grows, it can become the dominant contribution governing the departure from the GR-only solution in the regime where that organized response becomes significant. This does not remove or replace the Einstein curvature term. In black-hole physics specifically, GR remains the controlling description for the exterior spacetime, the trapping horizon, orbital dynamics, and the ordinary relativistic evolution across most of the solution. The polarization–transport components become dynamically significant only in the extreme compact regime where the GR-only continuation would otherwise approach an infinite-curvature endpoint that is not physically admissible in CPTG. The reverse transition is equally continuous: where polarization and transport weaken, the solution moves smoothly back toward the GR-dominated limit of Eq. (9). There is therefore no discontinuous transition from one gravity theory to another.

The physical interpretation is correspondingly conservative: CPTG preserves the Einsteinian identification of gravity with geometry. General Relativity is the Einstein component of the coupled gravitational response, while curvature polarization and curvature transport provide additional geometric organization where their contributions become significant. These components operate together within one gravitational framework rather than as replacements for one another. GR dominates wherever the polarization–transport contribution is negligible. For a black hole, the event or trapping horizon is not itself a boundary at which polarization–transport terms become significant: the horizon and most

exterior strong-field physics remain GR-controlled, while the polarization–transport terms become important only in the deeper regime where a GR-only continuation would otherwise run to an inadmissible infinite-curvature endpoint. When the polarization–transport contribution again becomes negligible, the ordinary Einstein equation is recovered.

Equation (9) is the exact formal recovery statement. Precision recovery in a system where GR is already tightly tested is a separate empirical requirement: after the standard source and ephemeris parameters are treated consistently, any residual CPTG polarization or transport signal must remain below the relevant observational limits [10, 11]. This distinction prevents the existence of the GR limit from being mistaken for a claim that every precision-limit test has already been completed.

The contracted Bianchi identity constrains the same unified field equation. If the ordinary material source is independently conserved,

$$\nabla^\mu T_{\mu\nu} = 0, \quad (11)$$

then Eq. (8) requires

$$\nabla^\mu (\Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}}) = 0. \quad (12)$$

This is a parent-level consistency condition on the combined polarization–transport response generated by the action. It requires conservation of the sum of the two geometric responses but does not require the polarization and transport tensors to be separately conserved.

The Noether requirement also fixes the status of any auxiliary geometric fields used to construct the response. Let Ψ_A denote any independent direction, state, or order field appearing in $S_{\text{pol}} + S_{\text{trans}}$, with Euler–Lagrange residual

$$\mathcal{E}_A \equiv \frac{1}{\sqrt{-g}} \frac{\delta(S_{\text{pol}} + S_{\text{trans}})}{\delta\Psi_A}. \quad (13)$$

For a diffeomorphism-invariant parent response action, the associated Noether identity reduces to Eq. (12) when every independent auxiliary field is on shell, $\mathcal{E}_A = 0$, or when the apparent auxiliary object is instead constructed covariantly from the solved geometry and therefore is not an independent background field. Consequently, a transport direction may be held fixed only after a declared reduction. At parent level a prescribed external S^α would not by itself provide the claimed automatic Noether closure; the direction must be geometry-selected or dynamically constrained. The same rule applies to any spacetime-varying suppression or state field. Structure-specific reduced quantities such as $a_\star$ and κ_{red} may enter as derived solution scales or boundary/reduction data, but they are not universal parent couplings that can be assigned arbitrarily as coordinate-dependent backgrounds.

Equations (7)–(13) close the parent conservation logic. Their arbitrary-geometry component realization depends on the chosen covariant geometric fields; the expressions below are the reduced representatives used in the declared physical sectors.

A limited covariant bridge is available in a static region where a hypersurface-orthogonal

Killing field remains timelike. For the normalized static congruence, the proper acceleration is the spatial gradient of the Killing lapse, and in the weak-field limit its acceleration and projected gradient reduce to the Newtonian field and its spatial gradient. Consequently a static covariant scalar can reproduce the reduced field-plus-gradient structure

$$\mathcal{C}_{\text{stat}} = c^4 \left[\mathbf{a}_\mu \mathbf{a}^\mu + \kappa_{\text{stat}}^2 (D_\mu \mathbf{a}_\nu) (D^\mu \mathbf{a}^\nu) \right] \longrightarrow g_j g_j + \kappa_{\text{stat}}^2 (\partial_i g_j) (\partial_i g_j) \quad (14)$$

in the declared static weak-field regime. This bridge is included only to show that the reduced field-plus-gradient structure is not intrinsically tied to Cartesian notation. It is not used as a black-hole interior invariant and is not extended through a Killing horizon, where the normalized static congruence is no longer a regular global description. For a static high-gradient scalar reduction with $a_\star > 0$, define

$$C_{\text{red}} = g_j g_j + \kappa_{\text{red}}^2 (\partial_i g_j) (\partial_i g_j), \quad \chi_{\text{red}} = \frac{C_{\text{red}}}{a_\star^2}. \quad (15)$$

Here κ_{red} is the gradient length selected by the declared reduction; it is not a universal identification of the galaxy-fixed κ_{struct} with the compact-sector κ_c . The corresponding polarization representative is

$$\mathcal{L}_{\text{pol}}^{(\text{red})} = \beta a_\star^{2/3} C_{\text{red}}^{1/3} = \beta a_\star^{4/3} \chi_{\text{red}}^{1/3}. \quad (16)$$

For a declared organized transport direction S^α and a sector-appropriate suppression function Σ_{red} , the explicit directional transport representative is

$$\mathcal{L}_{\text{trans}}^{(\text{red})} = -\frac{\lambda_t}{2} (S^\alpha \Sigma_{\text{red}} \nabla_\alpha \Phi)^2. \quad (17)$$

Equations (16) and (17) expose the shared reduced variational content: polarization is gradient-sensitive and transport is directional and suppression-weighted. The quantities κ_{red} , S^α , Σ_{red} , and the effective normalizations that appear after reduction take the forms registered for the physical state, symmetry, and scale of the problem.

Physical state, symmetry, scale, and boundary data determine the operative sector reduction and its variables. Downstream observational fits test that reduction; they do not retrospectively redefine the assumptions or authority of the upstream construction.

2.2.1 Coupling and Transport-Direction Status

The transport direction S^α is a normalized geometric direction field, $S^\alpha S_\alpha = 1$, in the reduced sectors where an explicit direction is retained. Consistent with Eq. (13), it may be held fixed during a declared reduced variation only after the physical reduction has selected it. At parent covariant level it must either be constructed from the solved geometry/state or be promoted to an independent geometric field with its own equation of motion and normalization constraint. It is not an added matter four-vector and it is not assumed to be one universal preferred direction shared by every system. Different geometries may select different effective transport directions; a reduction must state the direction or the state information from which it is obtained.

The parent couplings also have a definite dimensional status. In the SI convention used here, with each Lagrangian density measured as an energy density, $[C_{\text{red}}] = \text{m}^2 \text{s}^{-4}$, $[\nabla_\alpha \Phi] = \text{m s}^{-2}$, and dimensionless S^α and Σ_{red} , dimensional consistency requires

$$[\beta] = \text{kg m}^{-7/3} \text{s}^{2/3}, \quad [\lambda_t] = \text{kg m}^{-3} \text{s}^2. \quad (18)$$

These are parent-action couplings, not extra matter densities. In reduced galaxy calculations their normalization is absorbed into the effective response, whereas directional merger or compact reductions may retain the corresponding coupling structure explicitly. The distinction reflects the sector reduction rather than a change of theory.

The distinction between polarization and transport is physical. Curvature polarization answers the local source–response question: how does curvature reorganize in response to the strength, gradient, and structure of the baryonic source? Curvature transport answers the subsequent dynamical question: how is that organized response carried, retained, redistributed, or directionally displaced through an extended or changing system? In a quasi-static system polarization can generate enhanced support while preserving recovery of the high-acceleration baryonic limit. When the geometry is approximately radial and slowly evolving, much of the transport sector can reduce to a structural remnant; when the geometry is strongly anisotropic or dynamical, as in a dissociative cluster merger, transport can remain explicit and shift the curvature-supported observable response relative to the collisional baryonic gas.

Observational projection is downstream of physical curvature transport. Once a native or reduced CPTG state is solved, a CMB, baryon acoustic oscillation (BAO), abundance, lensing, or other comparison map expresses that state in the required reporting coordinate without supplying an additional physical transport mechanism. Cosmological projections therefore remain responses of one native history, while the nuclear source geometry is propagated by the native reaction network without replacing its microscopic rates or dynamics.

2.3 Known Cross-Scale Realizations

The known CPTG realizations and currently identified scale- or realization-specific extensions can be summarized without implying an exhaustive branch taxonomy or requiring one reduced equation to be copied into every regime. Each row below identifies the physical content retained in a developed realization or the structural role of an identified extension within the common curvature–polarization–transport framework:

Galaxy	quasi-static baryonic source $\rightarrow$ nonlinear curvature polarization, high-acceleration suppression, and a radial residual of transport; $a_\star$ is inferred for the solved structure and R_c is derived.
Dissociative merger	anisotropic source geometry $\rightarrow$ polarization plus an explicit directional transport operator along the retained S^α .
Compact structure	GR-controlled exterior and ordinary relativistic dynamics $\rightarrow$ localized high-gradient polarization–transport response in the extreme compact regime, with fractional-curvature polarization softening, a bounded/suppressed transport admissibility condition, finite physical admissibility, and a structure-dependent solution family.
Cosmology	universe-scale curvature/transport state $\rightarrow$ one native π -branch, from which observational response coordinates are projected at common geometric phase.
Galaxy cluster	extended bound-system baryonic source geometry, constrained by observed rotational, dispersion, thermal, or mixed support $\rightarrow$ a cluster-scale bound-system polarization–transport solution to be derived within the common structural hierarchy rather than introduced as a separate gravity law.
Nuclear reactions	shared CPTG scalar response geometry plus exact conserved reaction-state directions $\rightarrow$ network-native dynamical pushforward in the light-nuclide state space.

These entries summarize the CPTG realizations developed here together with identified scale- or realization-specific extensions. They are not exhaustive and do not imply that every regime must be expressible through one common reduction; CPTG retains the curvature–polarization–transport framework while allowing the operative variables and reduced equations to follow the physical scale, symmetry, state, and boundary data of the problem.

For calculations, the physical regime must be specified before its reduced nonlinear equations are solved. The operative direction is the hierarchy in Eq. (1): source geometry and state data are fixed first, the symmetry/scale/state realization is declared second, the corresponding polarization–transport dynamics or native pushforward is solved third, and only then is the solved state translated into a reported observable or network coordinate.

This ordering prevents a convenient scalar formula from being mistaken for the framework as a whole. Compatibility is assessed within each declared realization by its mathematical consistency, physical domain, boundary/state assumptions, and consistency with the CPTG curvature–polarization–transport architecture; no exhaustive taxonomy is required.

Once this distinction is made, quantities must be distinguished according to whether they belong to the native theory or arise only after reduction, projection, conditioning, or numerical validation.

2.4 Native Geometry, Reduced Laws, and Comparison Coordinates

CPTG uses a strict hierarchy of quantity classes because the same symbol can otherwise acquire different physical meanings when it is translated between theory and observational pipelines. The classification is summarized in Table 1.

Class	Meaning	Preferred claim language
Native invariant	Quantity belonging to the defining CPTG geometric structure before observational conversion	fixed / defines
Exact derived quantity	Algebraic or integral consequence of native quantities	derived / follows
Reduced physical law	Regime-specific equation obtained after declared symmetry, scale, or state reduction	reduces to / is represented by
Projected coordinate	CPTG result mapped into the coordinate used by an observational or numerical sector	maps to / is projected to
Data-conditioned coordinate	Quantity whose normalization or endpoint explicitly inherits an external observational scaffold	conditioned on / constructed from
Frozen comparison input	Upstream numerical setting held fixed before a declared comparison	held fixed / supplied as input
Validated result	Numerical or observational test outcome inside a stated domain and gate	passes / is reproduced
Diagnostic or open quantity	Candidate relation or comparison object lacking full independent derivation or validation	indicates / remains open

Table 1: Parent-level CPTG quantity classes. The class fixes the strongest permissible claim language and prevents a projected or data-conditioned quantity from being promoted into the native theory.

This authority hierarchy also determines the validation sequence. The operational rule is

$$\begin{aligned}
 \text{derive the relation} &\longrightarrow \text{assign its authority class} \\
 &\downarrow \text{freeze the upstream specification for the declared test} \\
 &\downarrow \text{project or reduce as required} \\
 &\longrightarrow \text{compare with the independent target.}
 \end{aligned} \tag{19}$$

“Freeze” here is a validation rule, not a statement that CPTG is permanently unrevisable. It means that a failed downstream comparison cannot be rescued within the same declared test by retrospectively changing the upstream law, changing its authority class, or inserting a sector-specific hidden component. A revised upstream theory may of course be proposed, but it constitutes a new derivation and a new test rather than a repair of the original validation.

Because several CPTG reductions contain conspicuous rational or π -based constants, their provenance must be stated separately from their numerical success. Table 2 records whether each major upstream quantity is a native definition, an algebraic consequence, a registered reduced-law coefficient, or a frozen comparison coordinate. A fixed coefficient

Quantity	Origin / defining relation	Derivation status in this parent paper	Authority
$p_C, \mathcal{B}_\pi, \mathcal{T}_\pi$	$p_C = \pi, \mathcal{B}_\pi = 1 - 1/\pi, \mathcal{T}_\pi = 1/\pi$	p_C is the native curvature exponent; the two weights are exact algebraic complements	[18]
$a_{\pi,0}$	$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2}$	Native universe-scale anchor; H_∞ , $H_0^{(\pi)}$, and $R_{C,0}$ are derived from it	[18]
$p_{ac}, \mathcal{G}_T$	$p_{ac} = 3 - \pi/100, \mathcal{G}_T = p_{ac}/p_C$	p_{ac} is the fixed acoustic-projection exponent; $\mathcal{G}_T$ is its exact projection ratio	[18, 21]
η_{10}^{ac}	$\Omega_b h_{ac}^2 = 0.0225278574944$ with the fixed-temperature conversion $\eta_{10} \simeq 273.9 \Omega_b h^2$	Frozen acoustic baryon-to-photon comparison coordinate; it is upstream of the nuclear endpoint tests	[18, 22]
$N_{\text{eff}}^{\text{CPTG}}$	$N_{\text{eff}}^{\text{CPTG}} = p_{ac} = 3 - \pi/100$	Effective relativistic-degree comparison coordinate used by the primordial radiation mapping; it is not an additional native radiation species	[24]
Galaxy source weights	27/50, 23/50, 16/25 in Eq. (41)	Registered fixed coefficients of the galaxy reduced law in this parent paper; their authority here is the declared galaxy reduction rather than a universal parent invariant; gas+disk sum to unity, the bulge factor is $(4/5)^2$, and no per-galaxy refit is performed	Registered here
Galaxy suppression exponents	123/50 and $-11/50$ in Σ_g	Registered fixed exponents of the nonlinear galaxy reduced law in this parent paper; their authority here is reduction-specific rather than universal across CPTG scales or realizations, and they are not inferred separately for each galaxy	Registered here
Galaxy transport amplitude	$\eta = 1/2$ in Eq. (32)	Registered fixed coefficient of the radial residual-transport term in the galaxy reduced law in this parent paper; it is reduction-specific, is not a universal parent coupling, and is not an additional galaxy fit parameter	Registered here

Table 2: Provenance and derivation status of major upstream fixed quantities. The table distinguishes native and exact algebraic quantities from registered reduced-law coefficients and frozen comparison coordinates; reduction-specific coefficients are not promoted to universal parent invariants by successful downstream use.

is not promoted to a first-principles derivation merely because it is reused successfully downstream.

For completeness, the remaining fixed relations used below retain equally explicit authority boundaries:

- $H_{0,ac}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T}$ is the registered present-normalization projection rule used to construct the acoustic rate coordinate. The shorthand $H_{\text{transported}} = H_{\text{raw}} \sqrt{\mathcal{G}_T}$ refers to this normalization map, not to a point-by-point transformation of the full rate history [18, 20].
- $\mathcal{D}_c, \mathcal{A}_s$, and $\mathcal{T}_\tau$ are registered deterministic CMB comparison-map gates. They act

on frozen conventional likelihood coordinates and are not native CPTG matter or radiation parameters [18].

- The displayed reduced growth/lensing pair $(\mu_{\text{CPTG}}, \Sigma_{\text{CPTG}})$, together with $\mathcal{A}_{\text{lens}}^{\text{CPTG}}$, defines representative comparison-response coordinates rather than a unique native perturbation law [18].
- J_{CPTG} and K_{CPTG} are fixed analytic scalar inputs of the nuclear source construction; the network response tests them downstream rather than fitting them [22].
- $\tau_7 = \ln \pi$ is the fixed integrated mass-seven abundance-transport law. Its live-network realization is tested after the gate has been specified [23, 24].

2.5 Structural Acceleration Scale $a_\star$

The CPTG structural acceleration scale $a_\star$ is the nonnegative acceleration scale associated with the structure of a mass-bearing gravitating object, from small to large physical scales,

$$a_\star \in [0, \infty). \quad (20)$$

It is not a universal acceleration constant and it is not tied to one observational interface. Depending on the physical realization, the structural acceleration scale may be inferred from resolved dynamics, derived within a structural reduction, selected as part of a boundary-value solution, or remain observationally unresolved. These are different methods of determining the same CPTG quantity, not different definitions of $a_\star$. Its value is tied to the structure of the object rather than assigned universally from outside that object.

For a dynamically evolving gravitating system, the structural acceleration scale may be written $a_\star(t)$. This notation does not define a second CPTG quantity; it denotes the time-dependent realization of the same object-dependent structural scale as the system's mass distribution, kinematic state, and geometric organization evolve. In quasi-static reductions such as the present galaxy calculation, $a_\star$ represents the structural value associated with the solved observational state. In explicitly dynamical problems, $a_\star(t)$ may instead belong to the evolving solution. Its time dependence is determined by the object's structural evolution and is not assumed to follow the cosmological Hubble rate or any universal external clock.

In the galaxy regime, the structural acceleration scale plays an acceleration-scale role analogous to the MOND constant a_0 , but the theories make different assignments: MOND fixes a_0 as one universal constant, whereas CPTG permits the structure of each gravitating object to select its own $a_\star$ [4]. The analogy is therefore one of dynamical role and dimension, not an identification of laws, interpolation functions, or numerical values. The universe-scale quantity $a_{\pi,0}$ introduced later is a separate native anchor of the cosmological π branch and is not the structural acceleration scale $a_\star$ of a gravitating object.

For the positive branch, $a_\star > 0$, the associated curvature-transport radius is

$$R_c = \frac{48\pi c^2}{a_\star}. \quad (21)$$

The associated structural time scale is

$$\tau_{\star} = \frac{c}{a_{\star}}, \quad (22)$$

so that the dimensional inverse-time scale is simply $\tau_{\star}^{-1} = a_{\star}/c$. When the same structural geometry is written in the Hubble-equivalent form

$$R_c = (16\pi)^2 \frac{c}{H_{\text{struct}}}, \quad (23)$$

Eqs. (21) and (23) fix

$$H_{\text{struct}} = \frac{16\pi}{3} \frac{a_{\star}}{c}. \quad (24)$$

The zero endpoint is defined by the continuous structural limit rather than by direct division by $a_{\star}$:

$$a_{\star} = 0 \quad \implies \quad R_c \rightarrow \infty, \quad \tau_{\star} \rightarrow \infty, \quad H_{\text{struct}} \rightarrow 0. \quad (25)$$

Accordingly, dimensionless coordinates normalized by $a_{\star}$, such as $C/a_{\star}^2$ or $g/a_{\star}$, are positive-branch coordinates defined for $a_{\star} > 0$. When $a_{\star} = 0$, the physical state is evaluated with the corresponding unnormalized equation or its continuous endpoint. Thus allowing $a_{\star} = 0$ does not require assigning finite values to coordinates that explicitly divide by $a_{\star}$.

The structural acceleration scale $a_{\star}$ should also be kept distinct from Structural Mode N . The former is the object's structural acceleration scale; the latter is a post-solution measure of how the retained polarization–transport response is organized. A particular measurement method may constrain $a_{\star}$, but it does not redefine the quantity itself.

Thus $a_{\star}/c$ and H_{struct} have the same dimensions but are not the same defined quantity. Keeping them distinct avoids converting a dimensional analogy into an exact structural identity.

The cosmological branch will later introduce its own native acceleration and expansion normalization. Those quantities are not obtained by assigning an object's structural acceleration scale $a_{\star}$ to the Universe. Conversely, observational quantities such as CMB-inferred, luminosity-distance, lensing, or abundance coordinates do not become additional native branches merely because they carry units familiar from cosmology.

This hierarchy supplies a practical rule for the rest of the paper: first identify the physical regime and solve or derive the corresponding CPTG quantity; only then map it into the coordinates required by the observation or numerical network. The simplest complete example is a quasi-static disk galaxy, where the baryonic source is directly observable and the nonlinear curvature-polarization response can be solved as a radial field.

3 Bound and Dynamical Structure

3.1 Galaxy-Scale Curvature Polarization Reduction

The first complete bound-system reduction of CPTG is the quasi-static disk galaxy. This is the natural starting point after the general hierarchy above because the baryonic source is observationally resolved, the dominant field can be represented radially, and the curvature-polarization response can be solved directly against a measured rotation curve [1, 2]. In this regime explicit time dependence is neglected, approximate rotational symmetry is used, and the field is represented by a scalar potential

$$\mathbf{g} = -\nabla\Phi, \quad g = |\mathbf{g}|. \quad (26)$$

The reduced geometry retains both the magnitude and radial organization of the solved field through

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2. \quad (27)$$

The first term measures field strength. The second records how rapidly that field is reorganized with radius. The invariant is therefore not a mass-density profile and is not interpreted as hidden matter. It is a structural diagnostic of the baryon-sourced curvature response. In the registered galaxy reduction, κ_{struct} is held fixed at the one-kiloparsec gradient-length conversion; it is not inferred separately for each galaxy [12].

The galaxy-limit acceleration law combines four pieces: the mapped baryonic source, the leading curvature-polarization response, the radial remnant of curvature transport, and a nonlinear high-acceleration suppression. The displayed normalized suppression form is written on the positive branch $a_\star > 0$; the $a_\star = 0$ state is its continuous baryonic endpoint. The operative equation on the positive branch is

$$g(r) = g_{\text{bar,CPTG}}(r) + \frac{\sqrt{a_\star g_{\text{bar,CPTG}}(r)} + g_{\text{geom}}(r)}{\left[1 + (g(r)/a_\star)^{123/50} \right]^{11/50}}. \quad (28)$$

This equation is implicit because the total acceleration appears on both sides. The galaxy field is therefore obtained by solving a nonlinear fixed-point problem rather than by adding a prescribed halo profile to the baryonic curve. For interpretation, Eq. (28) may be written

$$g_{\text{CPTG}}(r) = g_{\text{bar,CPTG}}(r) + \Delta g(r), \quad (29)$$

with

$$\Delta g(r) = [g_{\text{pol}}(r) + g_{\text{geom}}(r)] \Sigma_g(r), \quad (30)$$

and

$$g_{\text{pol}}(r) = \sqrt{a_\star g_{\text{bar,CPTG}}(r)}, \quad \Sigma_g(r) = \left[1 + (g(r)/a_\star)^{123/50} \right]^{-11/50}. \quad (31)$$

The decomposition does not introduce additional fitted components. It only makes visible the different roles already contained in the solved equation: baryonic sourcing, polarization, residual transport, and nonlinear suppression.

The structural acceleration scale $a_\star$ has the status defined in Sec. 2.5. In the registered galaxy reduction it is inferred for each galaxy through a nested trial-and-solve procedure. For each positive trial value of $a_\star$, the corresponding transport radius $R_c(a_\star)$ is derived, the implicit CPTG field is solved over the observed radii, and the resulting rotation curve is compared with the data. The residual over those trial solutions determines the inferred structural acceleration scale for that galaxy. The transport radius is therefore not fitted independently, and neither $a_\star$ nor R_c is carried unchanged from one galaxy to another. The galaxy reduction therefore infers one structural acceleration scale $a_\star$ per galaxy: R_c is derived from it, the source weights and suppression exponents remain fixed, and Structural Mode N /CSMI is calculated only after the field solution is selected.

The radial transport remnant is

$$g_{\text{geom}}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad \eta = \frac{1}{2}, \quad (32)$$

and the corresponding curvature-transport radius is

$$R_c = \frac{48\pi c^2}{a_\star}. \quad (33)$$

For $a_\star > 0$, the registered radial transport profile is bounded for every trial or inferred structural acceleration scale. With $y = r/R_c$, define

$$h(y) = \frac{y^{7/5}}{1 + y^{12/5}}, \quad \frac{g_{\text{geom}}}{a_\star} = \eta h(y). \quad (34)$$

Its derivative is

$$\frac{dh}{dy} = \frac{y^{2/5} \left(\frac{7}{5} - y^{12/5} \right)}{(1 + y^{12/5})^2}, \quad (35)$$

so the unique maximum occurs at

$$\frac{r_{\text{peak}}}{R_c} = \left(\frac{7}{5} \right)^{5/12} \simeq 1.150500155. \quad (36)$$

Using $\eta = 1/2$, the exact normalized bound is therefore

$$0 \leq \frac{g_{\text{geom}}(r)}{a_\star} \leq \frac{5}{24} \left(\frac{7}{5} \right)^{7/12} \simeq 0.2535129312. \quad (37)$$

On the positive branch, the dimensionless profile and its bound are common to the registered galaxy reduction, but the structural acceleration scale $a_\star$ and radial scale R_c remain structure-specific. This is a bound on the reduced transport remnant, not a universal acceleration scale.

It is useful to express the same local scale family through the exact structural Hubble-equivalent rate

$$H_{\text{struct}} = \frac{16\pi}{3} \frac{a_\star}{c}, \quad (38)$$

for which

$$R_c = \frac{c}{H_{\text{struct}}} (16\pi)^2. \quad (39)$$

This relation is an internal structural identity of the local reduction. It should not be confused with the dimensional inverse-time quantity $a_\star/c$, nor with any observational Hubble endpoint. The native cosmological branch is introduced later through the distinct quantities $a_{\pi,0}$, H_∞ , and $H_0^{(\pi)}$.

The baryonic source field is constructed from the observed gas, disk, and bulge component curves before the nonlinear equation is solved. When a tabulated component carries a sign, the signed-square convention preserves its source orientation:

$$Q(V) = V|V|. \quad (40)$$

The registered CPTG source mapping is

$$V_{\text{bar,CPTG}}^2(r) = \left[\frac{27}{50} Q(V_{\text{gas}}(r)) + \frac{23}{50} Q(V_{\text{disk}}(r)) + \frac{16}{25} Q(V_{\text{bulge}}(r)) \right]_+, \quad [z]_+ \equiv \max(z, 0). \quad (41)$$

with

$$g_{\text{bar,CPTG}}(r) = \frac{V_{\text{bar,CPTG}}^2(r)}{r}. \quad (42)$$

The gas-disk pair satisfies

$$\frac{27}{50} + \frac{23}{50} = 1, \quad (43)$$

while the bulge coefficient is

$$\frac{16}{25} = \left(\frac{4}{5}\right)^2. \quad (44)$$

These coefficients are fixed elements of the registered CPTG source mapping; they are not refitted from galaxy to galaxy. The positive-part operation is the registered reduction's total-source domain projection after the signed component contributions have been combined; it preserves the signed-square handling at component level while ensuring the nonnegative centripetal source required by the square-root polarization response. It is an implementation rule, not an additional fitted quantity. For compact notation below, g_{bar} denotes this CPTG baryonic source unless another comparison source is explicitly identified.

3.1.1 Nested Structural-Scale Inference and Inner-Solve Well-Posedness

The registered galaxy calculation is a nested one-parameter structural inference. For one positive trial $a_\star > 0$, define

$$u = \frac{g}{a_\star}, \quad \zeta = \frac{g_{\text{bar}}}{a_\star}, \quad \xi = \frac{r}{R_c} = \frac{ra_\star}{48\pi c^2}. \quad (45)$$

Then the complete implicit galaxy law is

$$u = \zeta + \left[\sqrt{\zeta} + \frac{1}{2} \frac{\xi^{7/5}}{1 + \xi^{12/5}} \right] \left(1 + u^{123/50} \right)^{-11/50}. \quad (46)$$

Thus a change in trial $a_\star$ changes both the source coordinate ζ and the structural radius coordinate ξ ; R_c is not an independent second fit parameter.

Let $\mathcal{A}_{\text{gal}} \subseteq [0, \infty)$ denote the admissible structural-acceleration family for the galaxy inference. For each positive trial $a_\star > 0$, $R_c(a_\star)$ is derived first, Eq. (46) is solved at every observed radius, and

$$V_{\text{CPTG}}(r_i; a_\star) = \sqrt{g(r_i; a_\star) r_i} \quad (47)$$

is compared with the observed rotation curve through

$$\chi_{\text{CPTG}}^2(a_\star) = \sum_i \left[\frac{V_{\text{obs}}(r_i) - V_{\text{CPTG}}(r_i; a_\star)}{\sigma_{V,i}} \right]^2, \quad (48)$$

The endpoint $a_\star = 0$, when included in $\mathcal{A}_{\text{gal}}$, is evaluated with the unnormalized field equation rather than the positive-branch variables u , ζ , and ξ ; Eq. (55) below shows that this endpoint is the baryonic member of the same continuous trial family.

The inferred structural acceleration scale is therefore selected by

$$a_{\star, \text{gal}} \in \arg \min_{a_\star \in \mathcal{A}_{\text{gal}}} \chi_{\text{CPTG}}^2(a_\star). \quad (49)$$

This outer optimization need not have a unique interior minimum. What can be proved exactly is that its inner field solve is single-valued and numerically stable.

At fixed radius and positive trial $a_\star > 0$, define

$$b \equiv g_{\text{bar}} \geq 0, \quad A \equiv \sqrt{a_\star b} + g_{\text{geom}} \geq 0, \quad p \equiv \frac{123}{50}, \quad q \equiv \frac{11}{50}. \quad (50)$$

The corresponding root residual is

$$F(g) = g - b - A \left[1 + \left(\frac{g}{a_\star} \right)^p \right]^{-q}. \quad (51)$$

satisfies $F(0) \leq 0$, $F(g) \rightarrow +\infty$, and

$$\frac{dF}{dg} = 1 + \frac{Apq}{a_\star} \left(\frac{g}{a_\star} \right)^{p-1} \left[1 + \left(\frac{g}{a_\star} \right)^p \right]^{-q-1} > 0. \quad (52)$$

Hence every admissible positive trial branch has exactly one nonnegative physical acceleration at each radius, bounded by

$$g_{\text{bar}} \leq g \leq g_{\text{bar}} + \sqrt{a_{\star} g_{\text{bar}}} + g_{\text{geom}}. \quad (53)$$

The registered fixed-point map is a contraction on its invariant interval: the exact bounds are $\sup |\mathcal{M}'| < 0.292$ for $0 \leq \zeta \leq 1$ and $\sup |\mathcal{M}'| \leq 0.679$ for $\zeta \geq 1$. Therefore the registered initialization converges to that unique root rather than selecting among multiple local branches.

The solved family is also ordered by the trial structural acceleration scale. Both $\sqrt{a_{\star} g_{\text{bar}}}$ and the registered $g_{\text{geom}}(r; a_{\star})$ increase with $a_{\star}$, while the suppression factor increases with $a_{\star}$ at fixed g . Because $\partial F / \partial g > 0$, implicit differentiation yields

$$\frac{\partial g(r; a_{\star})}{\partial a_{\star}} > 0 \quad (54)$$

for every nontrivial response. Finally,

$$\lim_{a_{\star} \rightarrow 0^+} g(r; a_{\star}) = g_{\text{bar}}(r), \quad (55)$$

so the admissible endpoint $a_{\star} = 0$ is the baryonic member of the CPTG trial family. A numerical optimizer that works in $\log_{10} a_{\star}$ cannot represent zero directly; a solution driven to its lower positive boundary should therefore be compared with the explicit $a_{\star} = 0$ endpoint unless a positive interior minimum is independently resolved.

These results establish the correct hierarchy: outer inference of the structural acceleration scale $a_{\star}$, derived $R_c(a_{\star})$, a unique convergent inner solve for $g(r)$, and only then the downstream curvature-organization diagnostics. They do not turn $a_{\star}$ into a universal acceleration constant.

3.1.2 Limiting Behavior of the Galaxy Response

The registered galaxy reduction consists of the fixed source map, inferred structural acceleration scale $a_{\star}$, and nonlinear response law. Numerical implementation and residual-analysis details are given in [12, 31].

The limiting behavior of the nonlinear response explains why the same equation can describe both compact inner regions and low-acceleration outer regions. At high acceleration,

$$g \gg a_{\star} \quad \implies \quad \Sigma_g(r) \ll 1, \quad (56)$$

so the additional curvature response is suppressed and

$$g \approx g_{\text{bar, CPTG}}. \quad (57)$$

At low acceleration,

$$g \ll a_{\star} \quad \implies \quad \Sigma_g(r) \rightarrow 1, \quad (58)$$

and therefore

$$g \sim g_{\text{bar}} + \sqrt{a_{\star} g_{\text{bar}}} + g_{\text{geom}}(r). \quad (59)$$

The square-root term supplies the leading polarization enhancement when it dominates the two smaller contributions. This limiting behavior gives the galaxy-scale physical interpretation of the coupled CPTG hierarchy. In the higher-acceleration inner galaxy, the polarization–transport contribution is suppressed and the gravitational response remains close to the directly baryon-sourced limit. As the baryonic acceleration decreases toward the extended outer galaxy, the suppression relaxes: curvature polarization supplies the leading enhanced response through $\sqrt{a_{\star} g_{\text{bar}}}$, while g_{geom} retains the radial contribution of curvature transport. The resulting gravitational field can therefore support the observed outer rotational kinematics without introducing an additional dark-matter mass distribution. The material source remains the observed baryonic matter; the additional response is carried geometrically by curvature polarization and transport. That intermediate limit produces the baryonic Tully–Fisher scaling and provides the next natural reading of the solved galaxy field.

3.2 Galaxy Structure and the Baryonic Tully–Fisher Scaling

The baryonic Tully–Fisher relation (BTFR) is the first analytic consequence of the weak-field galaxy reduction. Its origin in CPTG is not a separately imposed empirical interpolation law; it is the leading scaling of the square-root polarization term under a specified intermediate-regime approximation.

Suppose the galaxy has entered a low-acceleration interval in which

$$g_{\text{bar}}(r) \ll a_{\star}, \quad g(r) \ll a_{\star}, \quad (60)$$

so that $\Sigma_g \simeq 1$, and suppose further that over the observed flat-curve interval

$$g_{\text{bar}}(r), |g_{\text{geom}}(r)| \ll \sqrt{a_{\star} g_{\text{bar}}(r)}. \quad (61)$$

Then Eq. (28) reduces at leading order to

$$g(r) \simeq \sqrt{a_{\star} g_{\text{bar}}(r)}. \quad (62)$$

For the intermediate flat-regime scaling argument, approximate the exterior baryonic source by an effective enclosed mass,

$$g_{\text{bar}}(r) \simeq \frac{GM_{\text{eff}}}{r^2}, \quad g(r) = \frac{V^2(r)}{r}. \quad (63)$$

Substitution into Eq. (62) gives

$$V_f^4 \simeq G a_{\star} M_{\text{eff}}. \quad (64)$$

Define

$$M_{\text{eff}} = \Xi M_b \quad (65)$$

where Ξ is the effective source-mapping factor associated with the declared baryonic approximation over the scaling interval; it is not an additional gravitating component and is not introduced as a universal fitted constant. This yields

$$V_f^4 \simeq G a_\star \Xi M_b. \quad (66)$$

Equation (66) therefore derives the fourth-power form of the weak-field scaling. An exact population slope is a stronger statement because the structural product $a_\star \Xi$ may itself vary systematically with baryonic mass. Writing $A_{\text{str}}(M_b) = a_\star(M_b) \Xi(M_b)$ gives

$$\frac{d \ln V_f}{d \ln M_b} = \frac{1}{4} \left[1 + \frac{d \ln A_{\text{str}}}{d \ln M_b} \right]. \quad (67)$$

Thus the exact quarter-power population slope follows when A_{str} has no systematic mass dependence over the population being compared; the fourth-power functional form itself follows directly from the polarization-dominated scaling [3].

The status of Eq. (66) is important. It establishes the fourth-power scaling of the polarization-dominated regime, while its normalization is controlled by the structural product $a_\star \Xi$. Because $a_\star$ is object-dependent in the registered galaxy reduction, a universal BTFR normalization follows only when the structural product $A_{\text{str}} = a_\star \Xi$ is constant across the population under comparison. The parent law therefore fixes the scaling structure and the condition for an exact common normalization.

The BTFR therefore compresses only one aspect of the galaxy solution: its leading weak-field amplitude scaling. Two galaxies can occupy a similar BTFR location while organizing curvature differently across radius. To expose that information, the solved field must be interrogated for its internal radial structure. This motivates the Structural Mode N .

3.3 Structural Mode N : Solved-Field Organization and CSMI

Structural Mode N is the principal post-solution structural coordinate of the current CPTG galaxy reduction. It is calculated from the retained CPTG field only after the structural acceleration scale $a_\star$, the derived transport radius R_c , and the nonlinear radial solution have been fixed. It is therefore not a rotation-curve fitting parameter, not a morphology variable, and not an externally assigned galaxy type. Its role is to measure where structural change in the solved curvature–polarization–transport response is concentrated [12].

For the continuous galaxy functional, define

$$N = \frac{R}{\lambda}, \quad (68)$$

where R is the outer solved radius and λ is a curvature-weighted organization scale. Using the invariant of Eq. (27),

$$w(r) = \left| \frac{dC_g}{dr} \right|, \quad (69)$$

the registered outer organization length is

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (70)$$

The interval $[R/5, R]$ and the fixed $\kappa_{\text{struct}} = 1$ kpc entering C_g are registered features of the present galaxy readout. They are applied identically across low- and high-mode galaxies and are not selected from bulge prominence, Hubble type, or any conventional morphology label. Whenever w is not identically zero on the window,

$$\frac{R}{5} \leq \lambda \leq R, \quad 1 \leq N \leq 5. \quad (71)$$

If $w \equiv 0$ throughout the window, the centroid denominator vanishes and N is undefined rather than assigned an arbitrary value.

The mode is invariant under an affine rescaling

$$C_g(r) \longrightarrow A C_g(r) + B, \quad A \neq 0, \quad (72)$$

because the common factor $|A|$ cancels from the centroid ratio. It also has an exact conditional radial-similarity invariance. If

$$\tilde{r} = sr, \quad \tilde{R} = sR, \quad \tilde{C}_g(\tilde{r}) = A C_g(\tilde{r}/s) + B, \quad (73)$$

with $s > 0$ and $A \neq 0$, then $\tilde{\lambda} = s\lambda$ and $\tilde{N} = N$. This is an invariance of the Structural Mode functional conditional on the stated transformation of C_g ; because the operative galaxy invariant contains the fixed physical scale $\kappa_{\text{struct}} = 1$ kpc, arbitrary physical resizing of a galaxy does not automatically imply Eq. (73). Thus N measures radial organization rather than the normalization, additive offset, or simple radial similarity of the solved invariant. Low N places the dominant structural variation farther outward in the registered domain; larger N places it progressively farther inward. The coordinate therefore measures polarization/transport organization, not morphology or bulge prominence.

The continuous functional in Eqs. (68)–(70) is the parent definition of Structural Mode. Any sampled-data estimator is a downstream numerical readout of that functional and must preserve its declared domain and weighting rule. The Curvature Structure Mode Index (CSMI) is likewise downstream of the solved value of N : it maps the continuous structural coordinate into a catalog layer but does not enter the field solution or the definition of Structural Mode itself. Numerical estimators, catalog thresholds, and population applications are specified in the dedicated Structural Mode study [12].

The parent inference direction is therefore

$$\text{solved CPTG field} \longrightarrow N \longrightarrow \text{downstream structural classification}. \quad (74)$$

3.4 Far-Outer Continuation and Slope Convergence

The far-outer continuation is a derived asymptotic law of the registered galaxy reduction under its fixed-outer-source continuation. Once the mapped baryonic velocity contribution is continued as a finite outer plateau,

$$V_{\text{bar,CPTG}}^2(r) \rightarrow \text{constant}, \quad (75)$$

Eq. (42) gives

$$g_{\text{bar}}(r) \propto r^{-1}. \quad (76)$$

In the low-acceleration far-outer regime where the square-root polarization response supplies the leading term,

$$\sqrt{a_{\star} g_{\text{bar}}(r)} \propto r^{-1/2}, \quad (77)$$

so the effective solution approaches

$$g(r) \propto r^{-1/2}. \quad (78)$$

Since $V^2 = gr$,

$$V(r) \propto r^{1/4}, \quad s_{\infty} \equiv \frac{d \ln V}{d \ln r} = \frac{1}{4}. \quad (79)$$

This far-outer law and the intermediate BTFR scaling arise from different source limits. The BTFR derivation uses an effective finite enclosed mass, $g_{\text{bar}} \propto r^{-2}$, while the fixed-outer-source continuation gives $g_{\text{bar}} \propto r^{-1}$. They are distinct consequences of the same galaxy response law under different declared asymptotic source conditions.

3.5 Coherent Galaxy Clusters as Extended Bound Systems

A coherent galaxy cluster is an extended bound system in the CPTG scale hierarchy. Its material source is the baryonic structure, while rotational support, velocity-dispersion support, thermal pressure, or a mixed support state supplies the dynamical interface required to constrain the cluster-scale gravitational field. The parent ordering is

$$\begin{array}{c} \text{cluster baryonic source} \\ \downarrow \\ \text{declared support relation} \\ \downarrow \\ \text{cluster-scale CPTG field solution} \end{array} \quad (80)$$

The operative cluster variables and structural scales are determined by the cluster reduction itself. Approximate coherence places the system in the extended bound-system hierarchy.

When the source becomes strongly anisotropic and dynamically separated, the reduction retains directional curvature transport explicitly.

3.6 Anisotropic Dynamical Curvature Transport

In a quasi-static approximately radial system, much of the transport sector can appear as a radial structural remnant. In a strongly anisotropic or dynamically separating system, the transport direction remains explicit. Polarization governs the nonlinear local source response, while transport governs persistence and directional redistribution of the organized curvature response.

For a declared transport direction S^α , define

$$\Psi_{\text{tr}} = S^\alpha \Sigma_{\text{tr}} \nabla_\alpha \Phi, \quad (81)$$

so that

$$\mathcal{L}_{\text{trans}}^{(\text{red})} = -\frac{\lambda_t}{2} \Psi_{\text{tr}}^2. \quad (82)$$

Varying with respect to Φ , after the physical reduction has selected the transport direction, gives the leading directional operator

$$\mathcal{O}_{\text{trans}}[\Phi] = \nabla_\alpha \left(\lambda_t S^\alpha S^\beta \Sigma_{\text{tr}}^2 \nabla_\beta \Phi \right), \quad (83)$$

with the additional nonlinear contribution from $\delta \Sigma_{\text{tr}}$ belonging to the fuller reduced variation. The tensor $S^\alpha S^\beta$ carries the directional organization, while Σ_{tr} controls retained or suppressed transport response.

Projection onto a dominant spatial direction $\mathbf{S}$ gives the schematic anisotropic reduced operator

$$\mathcal{K}_{\text{aniso}} = \nabla^2 \Phi - s_{\text{tr}} \nabla \cdot \left[\Sigma_{\text{tr}}^2 \mathbf{S} (\mathbf{S} \cdot \nabla) \Phi \right]. \quad (84)$$

Coherent bound systems may admit a support-based reduction in which transport is compressed into the solved structure, whereas anisotropic dynamical systems retain the directional transport operator explicitly. Both are reductions of the same parent polarization–transport architecture and obey the common-limit condition of Eq. (2).

3.7 High-Gradient Compact Structures

Compact objects probe the extreme high-gradient limit of the same CPTG curvature–polarization–transport framework. CPTG does not replace General Relativity across the black-hole solution: GR remains controlling for the exterior spacetime, the trapping horizon, orbital dynamics, and the ordinary relativistic strong-field regime. The compact CPTG reduction is restricted to the deeper regime in which a GR-only continuation would otherwise approach an infinite-curvature endpoint excluded from the CPTG physical solution class [13].

The compact reduction retains the structural invariant below; the normalized coordinate $\chi = C_c/a_\star^2$ is defined on the positive branch $a_\star > 0$:

$$C_c = g_j g_j + \kappa_c^2 (\partial_i g_j)(\partial_i g_j), \quad \chi = \frac{C_c}{a_\star^2}, \quad (85)$$

where $a_\star$ is the structural acceleration scale of the compact object and κ_c is the compact-sector gradient length. Neither quantity is imported from the galaxy or cosmological reductions. Their values belong to the compact structural solution and are not presently fixed by an exterior black-hole mass measurement alone.

The registered high-gradient polarization representative is

$$\mathcal{L}_{\text{pol},c} = \beta a_\star^{2/3} C_c^{1/3} = \beta a_\star^{4/3} \chi^{1/3}. \quad (86)$$

Its constitutive derivative with respect to the retained field gradient scales as $|\partial g|^{-1/3}$ when the gradient term dominates, so the formal high-gradient response softens rather than diverges. This is a property of the registered compact constitutive ansatz; it does not by itself supply a complete relativistic interior solution.

Transport is subject to the compact admissibility condition

$$0 \leq \Sigma \leq 1, \quad |\mathcal{L}_{\text{trans},c}| = \frac{\lambda_t}{2} (\Sigma X_c)^2, \quad X_c = |S^\alpha \nabla_\alpha \Phi|, \quad (87)$$

so bounded compact transport is a condition on the product ΣX_c rather than on Σ alone. The physical compact solution class is correspondingly restricted to finite-curvature states: an infinite-curvature endpoint is not an admissible physical CPTG black-hole solution.

The parent theory does not assume one universal compact interior. For a fully specified source state, symmetry class, structural acceleration scale, and boundary data, the completed strong-field equations must select a finite solution that matches continuously to the GR-controlled exterior. Different objects may therefore occupy different finite structural realizations. Present measurements of pre-existing black holes primarily constrain the exterior gravitational response and do not generally determine the physical core mass or internal $a_\star$ independently [13, 14, 15].

The compact literature also distinguishes the exterior gravitational-response mass from the physical core mass through

$$M_{\text{apparent}} = A_{\text{pol}} M_{\text{core}}, \quad M_{\text{geom}} = M_{\text{apparent}} - M_{\text{core}}. \quad (88)$$

This is retained here only as structural bookkeeping. The amplification coordinate A_{pol} is an output of the solved compact geometry; the parent paper does not assign a universal interior profile or infer A_{pol} from a single local gradient coordinate.

3.8 Pressure- and Mixed-Support Bound Systems

A pressure-supported or mixed-support galaxy does not require a new fundamental gravity law merely because its stars do not move on circular orbits. What changes is the observational interface between the measured kinematics and the gravitational field that must support them. The source remains baryonic and the reduced CPTG response remains a curvature-polarization/transport response, but the required acceleration can no longer be read directly from V^2/r alone.

3.8.1 Observed Support Coordinates

At effective radius R_e , a rotation-aware reduced second moment is

$$V_{\text{rms}}^2(R_e) = \sigma_0^2 + V_{R_e}^2, \quad (89)$$

with corresponding observed support acceleration

$$a_{\text{sup}}(R_e) = \frac{V_{\text{rms}}^2(R_e)}{R_e}. \quad (90)$$

In the pressure-dominated limit, the directly observed kinematic coordinate is

$$a_\sigma = \frac{\sigma_\star^2}{R_e}. \quad (91)$$

Neither a_{sup} nor a_σ is, by definition alone, the structural acceleration scale $a_\star$. They are observable support coordinates. A particular reduced calculation may use one of them as a boundary anchor or infer the corresponding structural acceleration scale, but that identification is reduction-specific and must not be promoted into a universal equality.

A spatially resolved pressure-supported system is governed observationally by the radial Jeans interface,

$$g_{\text{Jeans}}(r) = -\frac{\sigma_r^2(r)}{r} \left[\frac{d \ln \rho_\star}{d \ln r} + \frac{d \ln \sigma_r^2}{d \ln r} + 2\beta_J(r) \right], \quad (92)$$

where β_J is the orbital-anisotropy parameter. Equation (92) is the standard spherical Jeans interface used here as an observational dynamical relation, not as an additional CPTG field equation [8]. A full radial reduction reconstructs the baryonic source and the same gravitational field over radius and requires the solved field to support $\rho_\star(r)$, $\sigma_r(r)$, anisotropy, and any ordered rotation simultaneously. A one-radius reduction is the corresponding restricted support interface.

The parent theory retains the support coordinates and full radial Jeans interface required for an independently specified dynamical reconstruction.

3.8.2 Redshift and the Boundary Between Local and Cosmological Structure

High redshift does not by itself define a multiplicative transformation of a bound system's structural acceleration scale. CPTG therefore does not assume a universal proportionality between $a_\star$ and the cosmological Hubble rate. The structural acceleration scale $a_\star$ and observable support coordinates such as a_σ and a_{sup} are determined by the reduction and the system's baryonic and kinematic state.

Cosmological redshift does not by itself supply a separate bound-system reduction or rescale a local structural acceleration scale. When a bound-system analysis uses a cosmology-dependent observational convention, that convention remains downstream of the locally solved structure rather than defining its gravitational response. This is categorically different from imposing $a_\star \propto H(z)$ by dimensional analogy.

4 Native Cosmology

4.1 The Native Cosmological π Branch

The structural reductions developed above are local: a galaxy, compact object, merger, or pressure-supported system is solved from the baryonic source and the geometric state appropriate to that system. Cosmology requires a different starting point. The background itself must be specified before any observational channel can be projected from it. CPTG therefore begins its cosmological construction with one native geometric history, the π -branch [18]. Acoustic, luminosity, CMB, abundance, and other comparison quantities are introduced only after this native state has been fixed.

This ordering is essential. The native branch is not obtained by averaging observational Hubble determinations, by importing a matter inventory from another cosmological framework, or by fitting separate expansion laws to different datasets. It is the universe-scale geometric reduction of CPTG. Friedmann equations, a dark-matter component, and a cosmological-constant term are not used here as the governing definition of the branch. Conventional cosmological quantities may be used later as comparison coordinates, but they do not define the native CPTG background.

4.1.1 Geometric Partition and Native Expansion Law

The native branch is organized by the curvature exponent

$$p_C = \pi, \tag{93}$$

and by the complementary geometric weights

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad \mathcal{B}_\pi + \mathcal{T}_\pi = 1. \tag{94}$$

These quantities are geometric branch weights. They are not matter-density or dark-energy

fractions.

The normalized native expansion function is

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi} = 1 - \frac{1}{\pi} + \frac{1}{\pi} a^{-\pi}. \quad (95)$$

Because the weights sum to unity,

$$E_\pi(1) = 1, \quad (96)$$

so the present phase $a = 1$ is normalized without introducing an additional present-epoch fit parameter into the branch function itself.

The native cosmological rate is then

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a). \quad (97)$$

At large scale factor the transport term decays and the branch approaches

$$H_\pi(a \rightarrow \infty) = H_0^{(\pi)} \sqrt{\mathcal{B}_\pi} \equiv H_\infty. \quad (98)$$

At small scale factor the $a^{-\pi}$ term dominates. The same equation therefore contains both the earlier transport-dominated regime and the later geometric asymptote; they are limits of one native history rather than separately normalized cosmologies.

In redshift form, with $a = (1 + z)^{-1}$,

$$E_\pi(z)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi (1 + z)^\pi, \quad H_\pi(z) = H_0^{(\pi)} E_\pi(z). \quad (99)$$

4.1.2 Acceleration Anchor and Present Normalization

The universe-scale branch carries the native defining acceleration anchor

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2} = 5.568327996831708 \times 10^{-10} \text{ m s}^{-2}. \quad (100)$$

The dimensionful normalization in Eq. (100) is a defining native branch anchor: $\pi^{3/2}$ fixes its dimensionless geometric coefficient, while the displayed $10^{-10} \text{ m s}^{-2}$ sets the declared universe-scale dimensional normalization. Quantities such as H_∞ , $H_0^{(\pi)}$, and $R_{C,0}$ are then derived from that anchor rather than used to fit it.

This is not the structural acceleration scale $a_\star$ of a galaxy or compact object. The two quantities belong to different geometric states and must not be interchanged.

For the native cosmological branch, the anchor fixes the asymptotic rate through

$$H_\infty = \frac{a_{\pi,0}}{c} = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (101)$$

and therefore

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (102)$$

The dependency chain is consequently

$$a_{\pi,0} \longrightarrow H_\infty \longrightarrow H_0^{(\pi)} \longrightarrow H_\pi(a), \quad (103)$$

not an interpolation between external CMB and distance-ladder determinations.

Equation (101) is specific to the universe-scale native anchor. It must not be confused with the Hubble-equivalent identity associated with the structural acceleration scale

$$H_{\text{struct}} = \frac{16\pi}{3} \frac{a_\star}{c}, \quad (104)$$

which belongs to the scale family associated with the structural acceleration scale $a_\star$ developed earlier. The equality $a_{\pi,0}/c = H_\infty$ is therefore not a license to identify an arbitrary $a_\star/c$ with a cosmological Hubble rate.

4.1.3 Native Curvature-Transport Scale

The corresponding present native curvature-transport radius is

$$R_{C,0} = 48\pi \frac{c^2}{a_{\pi,0}} = 788.7823091477 \text{ Gpc}. \quad (105)$$

Here the capitalized R_C denotes the cosmological native curvature-transport scale. It is distinct from the local structural radius R_c used in bound-system reductions, and it is not an FLRW spatial-curvature radius, a photon mean free path, or a dark-matter scale.

To keep the local and universe-scale acceleration coordinates visually distinct, define the epoch-dependent native cosmological acceleration coordinate

$$a_\pi(a) \equiv a_{\pi,0} E_\pi(a). \quad (106)$$

The native radius then evolves as

$$R_C(a) = \frac{48\pi c^2}{a_\pi(a)} = \frac{R_{C,0}}{E_\pi(a)}. \quad (107)$$

The rate and radius are reciprocal expressions of the same branch state:

$$H_\pi(a) R_C(a) = H_0^{(\pi)} R_{C,0} = \frac{48\pi c}{\sqrt{\mathcal{B}_\pi}}. \quad (108)$$

Thus increasing native rate corresponds to decreasing native curvature-transport radius along the same geometric trajectory. No second background history is required to relate the two.

4.1.4 Geometric Phase and Epoch Labels

For the parent paper the common cosmological phase is written

$$N_\pi \equiv \ln a. \quad (109)$$

Dedicated cosmology papers often write this phase simply as $N = \ln a$; the subscript is retained here to avoid collision with the established galaxy Structural Mode N . Equal N_π means equal position on the native geometric trajectory.

Redshift and radiation temperature may label that same trajectory through the ordinary epoch relations

$$a = \frac{1}{1+z}, \quad \frac{T}{T_0} = a^{-1} = 1+z. \quad (110)$$

The temperature relation is an epoch translation, not a new CPTG state variable, not a replacement time coordinate, and not an independent thermal branch. Once the native trajectory is specified, z or T/T_0 simply selects the corresponding value of a , and therefore of E_π , H_π , a_π , and R_C .

4.1.5 Kinematic Behavior of the Native Branch

It is useful to describe how the single native history changes character without interpreting its geometric weights as material components. Define the instantaneous transport fraction of the branch by

$$f_\pi(a) = \frac{\mathcal{T}_\pi a^{-\pi}}{E_\pi(a)^2}. \quad (111)$$

Differentiating Eq. (95) gives

$$\frac{d \ln E_\pi}{d \ln a} = -\frac{\pi}{2} f_\pi(a). \quad (112)$$

A purely kinematic deceleration coordinate can therefore be written

$$q_\pi(a) \equiv -1 - \frac{d \ln H_\pi}{d \ln a} = -1 + \frac{\pi}{2} f_\pi(a). \quad (113)$$

This quantity summarizes the rate evolution of the native branch; it does not introduce a matter or dark-energy decomposition.

In the early transport-dominated limit,

$$q_\pi(a \rightarrow 0) \longrightarrow \frac{\pi}{2} - 1 = 0.5707963268, \quad (114)$$

whereas at the present phase

$$q_{\pi,0} = -\frac{1}{2}. \quad (115)$$

The branch crosses $q_\pi = 0$ at

$$a_{\text{tr}} = 0.6564607036, \quad z_{\text{tr}} = 0.5233204280. \quad (116)$$

The change from decelerating to accelerating expansion therefore follows from the native π -branch itself. No separate cosmological-constant term is required in the definition of Eq. (95).

4.1.6 Why the Native Branch Must Come First

The branch defined above is the common cosmological source state for every later CPTG comparison layer. It fixes the native phase, rate, acceleration coordinate, and curvature-transport scale before an acoustic, luminosity, CMB, abundance, or distance coordinate is constructed. This is the cosmological analogue of the hierarchy established earlier for bound systems: first solve or define the native geometric state, then form the reduction required by the physical problem.

The native branch remains a cosmological background construction rather than a generic rescaling law for local bound systems. A bound structure is solved from its own physical source, support state, and boundary data; redshift alone does not redefine its local structural acceleration scale or structural organization. The next section therefore turns directly to the downstream observational projections of the native cosmological branch.

5 Observational Projection Hierarchy

5.1 Common-Phase Projection Architecture

The native cosmological branch defined above is the source state of the cosmological construction. Observations, however, do not all interrogate that state through the same physical response channel. Acoustic propagation, luminosity-distance calibration, CMB parameterization, growth, and lensing each expose a different reduction or projection of the same underlying geometry. The purpose of the projection hierarchy is to translate between those channels without allowing the observational coordinates to redefine the native branch [18].

The architecture is therefore

$$\mathfrak{B}_\pi \longrightarrow \{\mathfrak{P}_{\text{ac}}, \mathfrak{P}_L\}, \quad (117)$$

where $\mathfrak{B}_\pi$ is the one native cosmological history and $\mathfrak{P}_{\text{ac}}$ and $\mathfrak{P}_L$ are its acoustic and luminosity projections. They are not independently initialized cosmologies and do not carry independent primordial native normalizations.

The shared geometric phase remains

$$N_\pi = \ln a. \quad (118)$$

For any defined projection b , its rate generates a projection-specific time coordinate through

$$\frac{dN_\pi}{dt_b} = H_b(N_\pi), \quad H_b dt_b = dN_\pi. \quad (119)$$

Thus equal N_π means equal geometric phase, not equal accumulated projection time. The native branch itself generates

$$t_\pi(a) = \int_0^a \frac{da'}{a' H_\pi(a')}, \quad t_{0,\pi} = 12.7541210323 \text{ Gyr}. \quad (120)$$

The corresponding acoustic and luminosity coordinates are introduced below from their projected rate histories. These integrated projection times are response coordinates generated by the declared rates. The parent theory does not infer from their numerical values that all projected clocks are independent physical proper-time standards.

The distinction between source state and response coordinate is easiest to see by isolating the nonconstant part of the native branch,

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}. \quad (121)$$

There is one such native background function. The first question is therefore not how to fit an acoustic history independently, but whether the acoustic state follows uniquely from X_π . It does.

5.2 Exact Acoustic Projection of the Native Branch

The acoustic sector is the cleanest demonstration of the projection principle because its background mapping is explicit. The acoustic exponent is

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464102, \quad (122)$$

and the corresponding curvature-to-acoustic projection ratio is

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{\pi} = 0.9449296585513721. \quad (123)$$

Define the local scalar projector

$$\mathcal{P}_{\text{ac}}(X) \equiv \mathcal{T}_\pi^{1-\mathcal{G}_T} X^{\mathcal{G}_T}. \quad (124)$$

Within the declared class of local monomial maps $\mathcal{P}(X) = AX^\gamma$, the two requirements that $\mathcal{T}_\pi a^{-p_C}$ map to exponent p_{ac} and that the coefficient $\mathcal{T}_\pi$ be inherited exactly force $\gamma = p_{\text{ac}}/p_C = \mathcal{G}_T$ and $A = \mathcal{T}_\pi^{1-\mathcal{G}_T}$. The projector is therefore uniquely fixed within that declared monomial class; “exact” below refers to the resulting algebraic mapping, not to a claim that no broader non-monomial representation could ever be written.

Applying it to the native nonconstant state gives

$$X_{\text{ac}}(a) = \mathcal{P}_{\text{ac}}[X_{\pi}(a)] \quad (125)$$

$$= \mathcal{T}_{\pi}^{1-\mathcal{G}_T} \left(\mathcal{T}_{\pi} a^{-\pi} \right)^{\mathcal{G}_T} \quad (126)$$

$$= \mathcal{T}_{\pi} a^{-p_{\text{ac}}}. \quad (127)$$

The acoustic background is therefore

$$E_{\text{ac}}(a)^2 = \mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-p_{\text{ac}}}. \quad (128)$$

The important result is not merely the new exponent. Equation (127) preserves the native coefficient $\mathcal{T}_{\pi}$ exactly. The acoustic response therefore inherits its primordial background normalization from the native state; there is no additional free acoustic amplitude that must be synchronized with a separately initialized native history. This is the mathematical reason CPTG treats the acoustic sector as a projection rather than a second cosmology [21, 18].

For Hubble-like projected rate coordinates, the fixed CPTG transport rule acts on the present normalization,

$$H_{0,\text{transported}} = H_{0,\text{raw}} \sqrt{\mathcal{G}_T}. \quad (129)$$

Applying this normalization rule to the native present rate gives the associated acoustic coordinate

$$H_{0,\text{ac}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (130)$$

This value is fixed by the native normalization and the projection ratio before any CMB comparison is performed. It does not imply that the full acoustic rate history is obtained by multiplying the native history by one constant. Defining

$$H_{\text{ac}}(a) \equiv H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac}}(a), \quad (131)$$

the relation to the native history is

$$\frac{H_{\text{ac}}(a)}{H_{\pi}(a)} = \sqrt{\mathcal{G}_T} \frac{E_{\text{ac}}(a)}{E_{\pi}(a)}, \quad (132)$$

which reduces to $\sqrt{\mathcal{G}_T}$ at $a = 1$ because $E_{\text{ac}}(1) = E_{\pi}(1) = 1$ [18, 20].

The acoustic projected time coordinate is generated by its own rate history,

$$t_{\text{ac}}(a) = \int_0^a \frac{da'}{a' H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac}}(a')}. \quad (133)$$

For the constant-exponent form above,

$$t_{0,\text{ac}} = 13.88517358 \text{ Gyr}. \quad (134)$$

This is an integrated acoustic response-time coordinate. It is not promoted here to a universal physical age or used to replace the native phase variable.

The acoustic projection establishes how one native background is expressed in the CMB-facing acoustic channel. The late distance-ladder problem asks the complementary question: how does the same native phase appear through a luminosity-distance response?

5.3 Luminosity Projection and the Hubble-Bridge Boundary

At fixed common phase, write the luminosity response in the general form

$$H_L(N_\pi) = \mathcal{L}(N_\pi)H_\pi(N_\pi), \quad \Xi_L(N_\pi) \equiv \frac{1}{\mathcal{L}(N_\pi)}. \quad (135)$$

Its projected time coordinate obeys

$$\frac{dt_L}{dt_\pi} = \Xi_L(N_\pi), \quad H_L dt_L = H_\pi dt_\pi = dN_\pi. \quad (136)$$

The luminosity response is therefore indexed by the same native geometric phase; it is not an independently initialized cosmological clock or a second native expansion history. This is the theoretical content required by the parent framework.

Unlike the acoustic projector, however, the present theory does not yet contain a microscopic independently normalized expression for $\mathcal{L}(N_\pi)$. Current Hubble-bridge realizations therefore use an external late-distance normalization and remain data-conditioned comparison constructions. Their endpoint values, integrated comparison times, and separation between the acoustic and luminosity coordinates are given in [20].

A predictive luminosity sector requires $\mathcal{L}(N_\pi)$ to be derived and normalized upstream of the distance-ladder endpoint. Once that is achieved, the luminosity response can be tested without creating permission for a second native CPTG branch.

5.4 Native Horizon Containment and Common Background Origin

Once the native state and exact acoustic projector are separated conceptually, the cosmological horizon question separates into two parts. The spatial question asks whether the physical region associated with the last-scattering sphere fits inside one native curvature-transport domain at the same phase. The projection question asks whether the acoustic background on that domain is uniquely inherited from the native state.

Let $z_\star$ denote the declared last-scattering comparison epoch. The acoustic comparison distance is

$$\chi_\star = \frac{c}{H_0^{(\pi)}} \int_0^{z_\star} \frac{dz}{E_{\text{ac}}(z)}, \quad (137)$$

where the dimensional prefactor remains the native normalization while the exact acoustic

projection enters through the integrand. This is the registered native-normalized acoustic comparison-distance definition; replacing $H_0^{(\pi)}$ in the prefactor by $H_{0,\text{ac}}^{\text{CPTG}}$ would define a different comparison coordinate. The physical radius of that future-observable comoving region at a common phase is

$$L_\star(a) = a\chi_\star. \quad (138)$$

The like-for-like native containment coordinate is therefore

$$\Lambda_R(a) \equiv \frac{L_\star(a)}{R_C(a)}. \quad (139)$$

A native background-containment solution at a chosen phase requires $\Lambda_R < 1$; the equality $L_\star = R_C$ defines the corresponding entry phase. Radius is compared with radius throughout, not a full diameter with one native radius.

The background-horizon statement follows only after containment is combined with the exact projector. Within the declared single-native-solution background construction, the connected native domain carries one background value at each common phase. For the local scalar projector of Eq. (124), a homogeneous native background satisfies

$$\nabla_i X_\pi^{(0)} = 0 \quad \implies \quad \nabla_i X_{\text{ac}}^{(0)} = \mathcal{P}'_{\text{ac}}[X_\pi^{(0)}] \nabla_i X_\pi^{(0)} = 0. \quad (140)$$

Thus the acoustic background inherits the native background state algebraically. CPTG does not require separately initialized acoustic patches to exchange heat across a region that is already represented as one connected native background domain. This is a common-origin statement, not a claim of superluminal material propagation, and it is a background-level closure rather than a substitute for a future perturbation-level causal-propagation calculation. The numerical containment ratio, entry phase, and associated distances are given in [21].

5.5 CMB, Growth, and Lensing as Comparison Coordinates

The native background does not by itself supply every conventional variable used by CMB and large-scale-structure likelihoods. CPTG therefore uses a translation layer. In this layer standard scalar-perturbation notation, effective Poisson relations, Weyl-potential combinations, density coordinates, and compressed amplitudes are comparison coordinates for expressing CPTG response in observationally familiar variables. They are not adopted as the governing FLRW equations of the native CPTG branch [18].

For example, using standard scalar-perturbation notation [9], denote the conventional comparison potentials by Ψ_{cmp} and Φ_{cmp} to avoid collision with the CPTG source potential Φ used earlier. A conventional metric interface is

$$ds^2 = -(1 + 2\Psi_{\text{cmp}})c^2 dt^2 + a(t)^2(1 - 2\Phi_{\text{cmp}})d\mathbf{x}^2. \quad (141)$$

The corresponding response functions are defined through

$$k^2 \Psi_{\text{cmp}} = -4\pi G a^2 \bar{\rho} \mu_{\text{CPTG}}(a, k) \delta, \quad (142)$$

and

$$k^2(\Phi_{\text{cmp}} + \Psi_{\text{cmp}}) = -8\pi G a^2 \bar{\rho} \Sigma_{\text{CPTG}}(a, k) \delta. \quad (143)$$

These equations define the comparison interface; they do not replace the parent curvature-polarization and transport field structure with a conventional dark-sector perturbation model.

5.5.1 Projection-Law Status

The comparison functions $\mu_{\text{CPTG}}(a, k)$ and $\Sigma_{\text{CPTG}}(a, k)$ are interfaces through which a solved CPTG cosmological response is represented in conventional growth and lensing coordinates. A CMB comparison map is likewise a downstream projection of the fixed native/acoustic construction into the harmonic or likelihood coordinate used by the observational analysis [19]. The parent law requires these projections to inherit the upstream CPTG state.

Sector-specific deterministic gates, frozen likelihood inputs, compressed amplitudes, and numerical map constructions are implementation-level realizations of this projection law and are given in the dedicated comparison-coordinate analyses [18, 19].

5.5.2 Authority Boundary of the Comparison Layer

The parent symbol $\mathcal{A}_{\text{lens}}^{\text{CPTG}}$ is a CPTG branch-response coordinate, not the conventional CMB nuisance parameter A_L . A single fixed CPTG row can be propagated through CMB, map-envelope, growth, and lensing diagnostics without redefining the native branch for each dataset; detailed bandpower, S_8 , map-envelope, survey, and likelihood translations are reported in the comparison-coordinate analysis [18]. The parent theory retains only the translation map required to connect those comparison coordinates to the unchanged native branch.

The parent claim is narrower and more important conceptually. The hierarchy is

$$\begin{aligned} \text{native CPTG geometry} &\longrightarrow \text{derived projection operators} \\ &\downarrow \text{frozen comparison inputs where required} \\ &\downarrow \text{derived observational coordinates} \\ &\longrightarrow \text{likelihood or survey tests.} \end{aligned} \quad (144)$$

A conventional variable that appears at the right-hand side of this chain does not thereby become part of the native CPTG matter inventory. Likewise, successful comparison-layer closure is not by itself a derivation of a complete native CPTG Boltzmann hierarchy or a full microscopic luminosity projector. Those remain stronger open problems.

With the cosmological projection hierarchy separated cleanly from the native branch, the theory can ask a more stringent question. If CPTG geometry is genuinely scalable,

can a similarly compact transport–polarization structure organize a microscopic reaction network without replacing that network’s native nuclear dynamics? The primordial nuclear sector provides that test.

6 Primordial Nuclear-Reaction Geometry

Primordial nucleosynthesis is a useful stress test of the framework because it is not a gravitational curve-fitting problem. A native Big Bang nucleosynthesis (BBN) code evolves many coupled nuclear species through reaction rates, thermodynamics, weak processes, and a changing radiation background. CPTG does not replace that machinery. Instead, it asks whether the *reaction-source directions* driving that high-dimensional system possess a lower-dimensional geometric organization before their endpoint abundance responses are examined. The dedicated nuclear-reaction theory develops and tests that construction in detail [22].

The resulting architecture mirrors the hierarchy already used elsewhere in the parent theory:

$$\begin{aligned}
 &\text{fixed CPTG geometric response} \longrightarrow \text{exact reaction-source geometry from stoichiometry} \\
 &\quad \downarrow \text{ network-native dynamical pushforward} \\
 &\quad \longrightarrow \text{observable abundance response.}
 \end{aligned} \tag{145}$$

The first two layers belong to the CPTG construction. The third belongs to the nuclear network. The fourth is what can be compared with numerical or observational endpoints. Keeping these layers separate is essential: a successful endpoint calculation is not allowed to redefine the source geometry that generated it.

6.1 What Is Shared with the Spacetime Theory

The nuclear construction does not claim that the four-dimensional reaction-source basis is obtained by reading four tensor components directly from $\Pi_{\mu\nu}^{\text{pol}}$ or $\Theta_{\mu\nu}^{\text{trans}}$. The shared CPTG content is more specific. The cosmological branch supplies the same geometric transport ratio $\mathcal{G}_T$ used by the primordial comparison coordinate, and the nuclear scalar source uses the fixed CPTG response coefficients J_{CPTG} and K_{CPTG} . Nuclear stoichiometry then supplies information that the spacetime action does not: exact baryon- and charge-conserving reaction directions in the light-nuclide state. The native network supplies the final dynamical susceptibility. Schematically,

$$\underbrace{(\mathcal{G}_T, J_{\text{CPTG}}, K_{\text{CPTG}})}_{\text{shared CPTG scalar geometry}} + \underbrace{\mathbf{s}_r}_{\text{nuclear stoichiometric source}} \xrightarrow{\mathcal{N}} \underbrace{\mathbf{y}_{r,\mathcal{N}}}_{\text{network-native response}} . \tag{146}$$

This separation is the precise spacetime-to-nuclear bridge used in the present theory. The shared scalar geometry constrains the source response; conservation laws determine the

nuclear source direction; and the network carries that source through its own dynamics. The four nuclear source coordinates are therefore a state-space realization of the same curvature–polarization–transport organization; they are not asserted to be literal tensor components of $\Pi_{\mu\nu}^{\text{pol}}$ or $\Theta_{\mu\nu}^{\text{trans}}$. The rank-four theorem follows exactly from the declared light-sector stoichiometry and conservation laws, then receives independent physical support from network-native pushforward tests [22]. The rank-four statement by itself is an exact stoichiometric/conservation result. Its CPTG-specific role begins when those conserved directions are coupled to the shared scalar geometry ($\mathcal{G}_T, J_{\text{CPTG}}, K_{\text{CPTG}}$) before the native network pushforward; stoichiometry alone does not establish the gravitational connection.

6.2 Cosmological Transport into the Primordial Coordinate

The nuclear construction is connected to the preceding cosmological projection hierarchy through the already-defined acoustic transport ratio

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C} = \frac{3 - \pi/100}{\pi} = 0.9449296585513721. \quad (147)$$

The acoustic baryon-to-photon reference coordinate is

$$\eta_{10}^{\text{ac}} = 6.170380167710, \quad (148)$$

and the transported CPTG BBN coordinate is

$$\eta_{10}^{\text{BBN}} = \eta_{10}^{\text{ac}} \sqrt{\mathcal{G}_T} \simeq 5.998071834744. \quad (149)$$

The displayed transported value is the retained stored-precision comparison coordinate. Reconstructing it only from the rounded acoustic decimal printed in Eq. (148) can differ in the last displayed digits.

Thus the primordial reaction calculation does not begin from a disconnected nuclear normalization. It receives a declared comparison coordinate from the cosmological transport hierarchy and then allows the native BBN network to perform the actual nuclear evolution. The distinction is the same one used throughout CPTG: the comparison coordinate specifies how the theory enters the external calculation; it does not turn the external calculation into CPTG geometry [18, 24].

At the scalar source level the fixed first- and second-order CPTG coefficients are

$$J_{\text{CPTG}}(\eta_{10}) = \frac{1}{\sqrt{8}} \sqrt{\frac{\eta_{10}}{\eta_{10}^{\text{ac}}}}, \quad (150)$$

and

$$K_{\text{CPTG}}(\eta_{10}) = -\frac{3}{16} \frac{\eta_{10}}{\eta_{10}^{\text{ac}}} \mathcal{G}_T^{-1/3} = -\frac{3}{2} J_{\text{CPTG}}^2 \mathcal{G}_T^{-1/3}. \quad (151)$$

These are fixed analytic inputs to the nuclear construction. The network response tests them without redefining the scalar coefficients in Eqs. (150) and (151) [22].

6.3 The Four-Sector Light-Nuclide State

The light-nuclear backbone is organized by structural level rather than by treating every reaction as an unrelated coordinate:

$$(n, p) \rightarrow D \rightarrow (T, {}^3\text{He}) \rightarrow {}^4\text{He}. \quad (152)$$

The corresponding interpretation is

$$\text{vertex} \rightarrow \text{bridge} \rightarrow \text{closure} \rightarrow \text{saturation}. \quad (153)$$

The vertex contains the free neutron–proton source population; deuterium is the first bound bridge; the tritium–helium-three mirror pair forms a charge-resolved mass-three closure sector; and helium four is the terminal saturated state of this declared light-element backbone. This ordering does not say that the physical network is a one-way four-reaction chain. Forward and reverse reactions remain fully active in the native code. The ordering instead supplies a state geometry in which the direction of each physical reaction can be represented [22].

The baryon-weighted dynamic state is

$$\mathbf{X} = \begin{pmatrix} X_V \\ X_B \\ X_C \\ X_S \end{pmatrix} = \begin{pmatrix} Y_n + Y_p^{\text{free}} \\ 2Y_D \\ 3(Y_T + Y_{{}^3\text{He}}) \\ 4Y_{{}^4\text{He}} \end{pmatrix}. \quad (154)$$

Coarse-graining the mirror sectors would lose physically important charge information, so CPTG retains the internal polarization coordinates

$$P_V = Y_p^{\text{free}} - Y_n, \quad P_C = 3(Y_{{}^3\text{He}} - Y_T). \quad (155)$$

Together, $(X_V, P_V, X_B, X_C, P_C, X_S)$ reconstruct the six species exactly; the corresponding linear coordinate transformation has determinant 288. The four-sector description is therefore a reversible change of coordinates for this declared six-species light sector once its polarization information is retained, not an abundance-destroying compression.

6.4 Ordered Transport and Charge-Constrained Polarization

Transport between successive structural sectors is represented by the incidence operator

$$\mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \frac{d\mathbf{X}}{dx} = \mathbf{D}\mathbf{j}, \quad (156)$$

where $\mathbf{j} = (j_{VB}, j_{BC}, j_{CS})^T$. Because

$$\mathbf{1}^T \mathbf{D} = 0, \quad (157)$$

the ordered transport operator preserves the baryon simplex exactly. The three columns of $\mathbf{D}$ therefore supply the vertex-to-bridge, bridge-to-closure, and closure-to-saturation transport directions.

Charge conservation supplies the missing internal direction. Within the same six-species sector, every baryon- and charge-conserving reaction perturbation satisfies

$$3 \delta P_V + \delta P_C = 0. \quad (158)$$

Only one independent polarization coordinate remains, which may be written

$$P_{\text{nuc}} = \delta P_V = -\frac{\delta P_C}{3}. \quad (159)$$

The universal reaction-source coordinate is consequently

$$\mathbf{s}_r = \begin{pmatrix} q_{VB} \\ q_{BC} \\ q_{CS} \\ P_{\text{nuc}} \end{pmatrix}_r. \quad (160)$$

This is the central structural result: three ordered transport coordinates plus one charge-constrained polarization coordinate give an exact rank-four reaction-source space in the declared light-element domain.

The rank follows directly from the structure above. The incidence matrix has three independent transport columns,

$$\text{rank } \mathbf{D} = 3, \quad (161)$$

and charge conservation leaves one independent polarization coordinate P_{nuc} . The reaction-source space therefore decomposes as

$$\mathcal{S}_{\text{nuc}} = \mathcal{T}_{\text{nuc}} \oplus \mathcal{P}_{\text{nuc}}, \quad \dim \mathcal{S}_{\text{nuc}} = 3 + 1 = 4. \quad (162)$$

This is the exact rank-four structural law of the declared light-element state space. Reaction-by-reaction bases and numerical validations are developed in the dedicated nuclear theory [22].

6.5 Network-Native Dynamical Pushforward

The exact source coordinate is universal within its declared domain, but its numerical evolution is not universal across nuclear codes. For network $\mathcal{N}$, let the first-order code-local

source drive be

$$\mathbf{d}_{r,\mathcal{N}}^{(1)} = \mathcal{A}_{r,\mathcal{N}}^{(1)} \mathbf{s}_r. \quad (163)$$

The first-order observable response is then represented by

$$\mathbf{y}_{r,\mathcal{N}}^{(1)} = \mathbf{L}_{\mathcal{N}}^{(1)}(\eta_{10}) \mathbf{d}_{r,\mathcal{N}}^{(1)}. \quad (164)$$

The matrix $\mathbf{L}_{\mathcal{N}}^{(1)}$ is network-native: each admissible nuclear network carries its own reaction rates, currents, Jacobians, source kernels, integration measures, and solver trajectory. The parent law fixes the exact CPTG source coordinate and then applies the native dynamical pushforward of the chosen network. Detailed numerical authorities, reaction surfaces, and cross-network validation belong to the dedicated nuclear papers [22].

6.6 Mass-Seven as a Separate Live Abundance-Transport Sector

Mass seven is intentionally kept outside the exact four-sector simplex. The light-sector theorem organizes the free-nucleon, deuterium, mass-three, and helium-four backbone; it does not acquire a fifth coordinate merely to address the cosmological lithium problem. Instead, the already-fixed CPTG mass-seven law acts directly on the live ${}^7\text{Li}$ and ${}^7\text{Be}$ network species [23, 24].

Define

$$Y_7 = Y_{7\text{Li}} + Y_{7\text{Be}}, \quad (165)$$

with live-channel sink

$$\left. \frac{dY_{7\text{Li}}}{dt} \right|_{\text{CPTG}} = \left. \frac{dY_{7\text{Li}}}{dt} \right|_{\text{native}} - \Gamma_7 Y_{7\text{Li}}, \quad (166)$$

$$\left. \frac{dY_{7\text{Be}}}{dt} \right|_{\text{CPTG}} = \left. \frac{dY_{7\text{Be}}}{dt} \right|_{\text{native}} - \Gamma_7 Y_{7\text{Be}}. \quad (167)$$

The integrated optical depth is fixed independently of the abundance endpoint,

$$\tau_7 = \int \Gamma_7(t) dt = \ln \pi, \quad e^{-\tau_7} = \frac{1}{\pi}. \quad (168)$$

The factor $e^{-\tau_7} = 1/\pi$ is the exact multiplicative attenuation generated by the declared sink for population exposed to that sink in isolation. In the live reaction network, however, mass seven is simultaneously produced, destroyed, and redistributed by native reactions while the sink is active. The final gated-to-baseline abundance ratio is therefore a network output and need not equal $1/\pi$ for an arbitrary temporal support of $\Gamma_7(t)$; the locked CPTG invariant is the integrated sink optical depth $\tau_7 = \ln \pi$, whose live-network realization is tested prospectively [24, 22].

The radiation-side CPTG comparison coordinate is

$$N_{\text{eff}}^{\text{CPTG}} = p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464. \quad (169)$$

This is an effective relativistic-degree comparison coordinate used in the primordial mapping; it does not introduce an additional native radiation species, and network-specific input translation is defined in [24].

The full primordial comparison tuple may therefore be written

$$\Theta_{\text{BBN}}^{\text{CPTG}} = \left(\eta_{10}^{\text{BBN}}, N_{\text{eff}}^{\text{CPTG}}, \tau_7 \right), \quad (170)$$

where each entry retains its own authority class [24, 22].

Extensions beyond the exact light-sector theorem are separate computational programs rather than additional axioms of the parent theory. Reduced-graph reachability and heavier-sector frontier studies are therefore cited but not reproduced here [25].

The nuclear sector thus supplies a distinct realization of the parent architecture: exact source geometry is fixed first, a native high-dimensional dynamical system performs the pushforward, and only afterward are endpoint observables examined.

7 Cross-Regime Consistency, Authority, and Falsifiability

The scale-structured framework is constrained by common parent-level consistency laws. Each physical realization retains its own variables and reduced equations, while its authority and relation to the common CPTG hierarchy remain fixed.

7.1 Cross-Regime Consistency Laws

The parent requirements are:

1. **Source and conservation consistency.** Gravitational reductions preserve the parent conservation condition of Eq. (12); non-gravitational state-space realizations preserve their native conservation laws.
2. **Scale declaration.** The source, physical scale, symmetry, dynamical state, and boundary or state data are declared before the operative reduction is solved.
3. **Overlap consistency.** Shared physical quantities agree in every genuine common domain according to Eq. (2).
4. **GR recovery.** Wherever the polarization–transport response becomes negligible, the gravitational solution approaches the Einstein limit of Eq. (9).
5. **Structural authority.** Object-dependent quantities such as a_* , derived scales such as R_c , and post-solution coordinates such as Structural Mode N retain the roles fixed by their defining reductions.
6. **Projection inheritance.** Acoustic, luminosity, CMB, growth, lensing, abundance, and other cosmological observational coordinates inherit one native cosmological state and remain downstream of that state.

7. **Native dynamical pushforward.** In the nuclear realization, CPTG fixes the source geometry while native reaction dynamics perform the evolution and preserve the applicable conservation structure.
8. **Finite compact admissibility.** A completed compact reduction remains finite in its physical interior and matches continuously onto the GR-controlled exterior in the common regime.

Violation of one of these requirements is a parent-level inconsistency of the corresponding realization.

7.2 Scope and Open Law Derivations

The principal law-level extensions are the explicit coherent-cluster bound-system reduction, completion of the finite compact boundary-value solution and its exterior matching law, and derivation of an independently normalized luminosity projector.

8 Conclusion

CPTG is a scale-structured geometric framework in which General Relativity, curvature polarization, and curvature transport operate as a coupled gravitational hierarchy. The common parent theory fixes source–response ordering, conservation, cross-scale matching, and the rule that each physical problem is represented by the reduction appropriate to its source, scale, symmetry, state, and boundary data.

In bound systems, the structural acceleration scale $a_\star$ belongs to the gravitating object, R_c is derived from that scale on the positive branch, and the nonlinear field is solved before downstream organization coordinates are constructed. Structural Mode N is a post-solution measure of the radial organization of the retained polarization–transport response. The weak-field galaxy reduction yields the BTFR-like fourth-power scaling in its declared intermediate regime and the $s_\infty = 1/4$ far-outer law under the fixed-outer-source continuation.

The same hierarchy extends by state and symmetry. Coherent galaxy clusters belong to the extended bound-system sector and are constrained through their baryonic source and support state. Strongly anisotropic dynamical systems retain the directional transport operator explicitly. Compact structures occupy the high-gradient sector, with finite physical admissibility and continuous matching to the GR-controlled exterior. Pressure- and mixed-support systems use the appropriate Jeans or second-moment interface to connect measured kinematics to the solved gravitational field.

Cosmology contains one native π -branch. Acoustic, luminosity, CMB, growth, lensing, abundance, and related observational coordinates are downstream projections of that native state at common geometric phase. The exact acoustic projector fixes the acoustic background from the native nonconstant state, while the luminosity sector is governed by its projection law pending an independently normalized microscopic projector.

The primordial nuclear sector realizes the same polarization–transport organization in a conserved reaction-state geometry. Three ordered transport directions and one charge-constrained polarization direction form an exact rank-four light-sector source space; native nuclear dynamics then perform the pushforward. The separate mass-seven law is fixed by the integrated transport depth $\tau_7 = \ln \pi$.

Across all scales, the defining requirement is consistency of the hierarchy: declare the physical realization, preserve its conservation laws, solve its operative dynamics, enforce agreement in genuine overlap limits, and only then construct downstream observational or numerical coordinates. This is the organizing law of CPTG as a framework of scales.

Data and Code Availability

The public CPTG repository contains the analysis software, benchmark implementations, companion manuscripts, and reproducibility materials used by the reductions summarized here: <https://github.com/CLG2025/CPTG>. External observational databases and numerical software retain their original provenance and citations.

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